
Victorian Education Infrastructure Investment and Household Travel Behaviour: A Narrative Visualisation

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1. Introduction

This narrative visualisation communicates three connected findings to Victorian policymakers and transport planners, derived from the Data Exploration Project (DEP). First, that Victoria's 2023–24 education infrastructure investment broadly targets outer-suburban growth corridors where household populations have concentrated, producing a moderate positive spatial alignment (Spearman $\rho = 0.597$, $p < 0.001$) between project counts and surveyed household density across 33 Local Government Areas (LGAs). Second, that the LGAs receiving the most investment are also the most car-dependent, with 89.1% of weekday work trips in High-investment areas made by private vehicle — not because investment causes car use, but because the outer-suburban geography of those same areas structurally limits access to public transport and active travel. Third, that the mechanism underlying this car dependency is a severe and measurable public transport time penalty: a median journey that takes 10 minutes by car takes 60 minutes by public transport in High-investment LGAs — a 6× multiplier that makes car use the only rational choice for time-constrained commuters.

The intended audience is Victorian government policymakers and transport planners — professionals who are data-literate, time-constrained, and need actionable findings rather than exploratory dashboards. This audience characteristic fundamentally shaped the design: findings must be pre-interpreted and sequenced, not left for the reader to self-assemble. The visualisation is designed to answer a question a policymaker would actually ask — not "what do the charts show?" but "where is our investment going, and what is it not fixing?" — and to conclude with a specific, measurable policy target rather than an abstract observation.

An important data limitation frames the entire visualisation. Of the 62 LGAs in Victoria with documented education investment, only 33 overlap with the VISTA 2024–25 household travel survey coverage area (Greater Melbourne and Geelong). The remaining 29 LGAs — predominantly in regional Victoria — have no available travel behaviour data. This gap is treated as a named finding rather than a methodological footnote: it is displayed prominently in the title panel and revisited in the LGA Explorer, because its policy implication (that the regions with the least data may face the worst transport access) is itself substantively important.

2. Design Process

The Five Design Sheet (FDS) methodology (Roberts et al., 2016) structured the design process from initial brainstorming through to a fully specified implementation plan. All five sheets are included in the Appendix.

2.1 Brainstorming and Idea Generation (Sheet 1)

Sheet 1 generated 26 candidate ideas spanning chart types, interaction mechanisms, layout patterns, and structural devices. Ideas were grouped into five functional categories: ideas that set context (e.g., headline metric cards, data coverage callout), ideas that frame the narrative (e.g., question decomposition, framing questions), ideas that show comparisons (e.g., scatter plot,

connected dot plot, slope chart), ideas that show distributions and variability (e.g., violin + boxplot, heatmap), and ideas that give the reader control (e.g., range slider, radio button, dropdown filter).

Eight ideas were filtered out at this stage with explicit justifications. The heatmap (idea 9) was rejected because, as Cleveland and McGill (1984) demonstrated, colour intensity is among the least accurate perceptual channels for encoding quantitative differences, making it unsuitable for communicating small percentage-point variations across transport modes. The stacked bar chart (idea 1) was set aside for trend comparison tasks on the grounds identified by Munzner (2014): when readers need to track an individual segment across bars, a floating baseline makes accurate comparison impossible. The diverging bar chart (idea 8) was rejected because no natural neutral midpoint exists in the investment or mode share variables — the chart form would impose a conceptual structure the data does not support. The linked highlight (idea 23) was noted but deprioritised as a standalone element because individual-row comparisons were less informative than tier-level patterns across the narrative. The scroll narrative (idea 13) was excluded due to JavaScript dependency constraints imposed by the course requirements, which mandate R Shiny rather than D3.

The most consequential decision at this stage was elevating the data coverage gap (idea 26) from a footnote to a structural element. Rather than disclosing in a methods section that 29 LGAs lacked travel survey coverage, the gap was positioned as part of the argument itself: the regions Victoria is investing in but cannot yet measure represent the most urgent knowledge gap for future transport planning decisions.

The brainstorming phase also identified three conceptually distinct design approaches for Sheets 2–4, differentiated at the level of narrative genre, perceptual task family, and interaction model — not merely at the level of layout. This distinction was made explicit in response to DVP Part 1 feedback and is documented in the Revision Record on Sheet 1.

2.2 Design A — Sequential Stepper (Sheet 2)

Design A is a fully author-controlled sequential narrative. The reader progresses through seven fixed steps — one chart, one analytical layer per step — using Next and Previous buttons. No filters or controls are available within the sequence; the only interactivity is hover tooltips for precise value retrieval. The step indicator at the top signals position within the narrative arc without allowing non-sequential navigation to unvisited steps.

The chart family in Design A is geographic and categorical: Step 2 uses a choropleth to establish spatial investment context, Step 3 uses a lollipop chart to show ranked LGA project counts, Step 4 uses a slope chart for mode share shift, Step 5 uses a grouped bar chart for vehicle ownership by income band, Step 6 uses a scatter plot for the SEIFA-to-mode-share bivariate relationship, and Step 7 presents the data coverage callout with a policy implication panel. This perceptual task family — geographic location, ranking, trend across an ordered axis, part-to-whole, and bivariate relationship — is entirely distinct from Design B (which centres on distributional shape and bivariate scatter) and Design C (which uses ranking, trend, and paired comparison).

The key strength of Design A is cognitive load management: one chart per step with one explanatory text panel prevents the reader from being pulled in multiple analytical directions simultaneously. Policymakers who are unfamiliar with the topic cannot skip to a chart before receiving the framing that makes it interpretable. The singular weakness is that comparative reading across steps requires memory rather than simultaneous visual access — a reader wanting to check whether the violin plot in Step 5 corroborates the slope chart in Step 4 must navigate back and forth.

2.3 Design B — Question-Answer Dashboard (Sheet 3)

Design B is a fully reader-controlled simultaneous dashboard. Three panels are permanently visible, each addressing a distinct question with a distinct encoding: a scatter plot (bivariate investment vs. household density), violin plots (distributional trip duration by tier), and small multiples (mode share by IRSD decile group). A left-side filter panel uses asymmetric scoping — the trip purpose dropdown affects the scatter and small multiples but not the violin plots, and the IRSD range slider affects the scatter and small multiples but not the violin plots, reflecting the analytical scope of each filter.

The chart family in Design B is distributional and bivariate: violin + boxplot for distribution shape with median and IQR, scatter for bivariate continuous relationship, and small multiples for faceted categorical comparison. No chart type is shared with Designs A or C.

Design B was ultimately rejected as the final implementation for three reasons documented on Sheet 3. First, showing all three questions simultaneously without guided entry risks a reader engaging only with a single panel and missing the connected argument. Second, the violin plots for trip duration across investment tiers produced a null result in the Kruskal-Wallis test for work trips (as established in the DEP), making that panel analytically weak as a centrepiece. Third, the asymmetric filter scoping — while analytically correct — requires reader orientation time that is incompatible with a time-constrained policymaker audience.

2.4 Design C — Martini Glass (Sheet 4)

Design C uses the Martini Glass structure described by Segel and Heer (2010) as a hybrid of author-driven and reader-driven narrative. The left column delivers a fixed three-panel guided narrative; the right column is locked behind an overlay until the reader explicitly triggers exploration mode. This structure preserves the sequential argument while rewarding readers who want to personalise the findings.

The chart family in Design C is ranking, trend, and paired comparison: a Cleveland dot plot (ranked) for investment-to-population alignment, a slope/connected line chart for mode share shift across tiers, and a connected dot plot (dumbbell) for paired car-versus-PT time comparison. These three encoding families are entirely distinct from the geographic/categorical family of Design A and the distributional/bivariate family of Design B, satisfying the FDS requirement for genuine structural distinctiveness across alternative designs.

The Martini Glass architecture directly addresses the audience constraint that motivated rejecting Design B: policymakers who are unfamiliar with the topic receive the full argument before controls become available, while experienced analysts can jump immediately to the LGA Explorer upon arrival. The one-way transition (guided narrative → exploration) enforced by the overlay ensures that early-exit readers still receive the complete argument without truncation.

One concern noted on Sheet 4 was that the simultaneous presence of the right column during State A (the locked narrative) could signal an exploration zone to experienced users before the narrative was complete, prompting premature engagement. This was addressed in the realisation by implementing a full-height sticky overlay with a blur backdrop on the right column, making its locked state visually unambiguous.

2.5 Realisation and Final Design (Sheet 5)

Sheet 5 specifies the implemented system. It synthesises elements across all three prior designs while making four deviations from Sheet 4 that are each justified on analytical and audience grounds.

Deviation 1: Filter zone removed from the right column. Sheet 4 included four filters (metric dropdown, trip purpose, investment tier checkboxes, IRSD range slider) in the exploration zone. These were removed from the final implementation. As documented on Sheet 5, filter-driven exploration places analytical burden on a policymaker audience that is inappropriate for the communication goal: requiring readers to manipulate four controls to assemble a conclusion actively contradicts the author-controlled framing that the Martini Glass structure is designed to deliver. The right column was redesigned as an LGA personalisation panel — a single dropdown that contextualises the already-delivered narrative findings for a specific suburb rather than enabling open-ended exploration. Removing the four-filter zone also reduces non-data interface elements that demand reader effort without adding analytical value — consistent with Tufte's (2001) principle that design should eliminate everything that does not serve the communication of the data itself.

Deviation 2: P1 redesigned as a Cleveland dot plot (ranked). Sheet 4 described P1 as a dot strip plot for investment dispersion across LGAs. The implemented P1 uses a ranked Cleveland dot plot with LGAs sorted ascending by household count on the Y-axis and project count on the X-axis, coloured by investment tier. This encoding change was made because the ranked layout, as recommended by Cleveland and McGill (1984), makes the alignment between household population and investment immediately readable through position on a common scale — the most accurate perceptual channel available (Cleveland & McGill, 1984). A conventional scatter plot with 33 labelled points would produce unacceptable label overlap; the ranked dot plot preserves all LGA identities while removing that problem.

Deviation 3: P2 redesigned as a slope/connected line chart. Sheet 4 used a bump chart for P2. This was revised to a slope chart with spline smoothing across the four investment tiers for two reasons. First, the bump chart idiom implies rank changes over time, which misrepresents the nature of the X-axis (investment tier is an ordered categorical grouping, not a temporal sequence). Second, a slope chart with labelled endpoints makes the direction and magnitude of mode share

shift across tiers immediately readable without requiring the reader to track rank positions — a task that a policymaker audience would find distracting relative to the substantive finding.

Deviation 4: P3 extended to two sub-charts by investment tier and IRSD decile. In Sheet 4, P3 showed the dumbbell comparison only by investment tier. The final implementation adds a second dumbbell below it showing the same car-versus-PT time gap across IRSD decile groups. This extension was necessary to address a potential confounding concern: that the PT penalty visible in the first dumbbell reflects investment tier rather than underlying geography and infrastructure. Showing the same penalty pattern across IRSD deciles — an independent dimension — confirms that the structural PT disadvantage is not an artefact of how tiers were defined.

Headline metric strip. A row of three headline statistics (15 projects in Casey; 89% of work trips by car; 6× longer by PT) was added to the title panel. This device serves Bertin's (1983) principle of visual hierarchy: before readers engage with the charts, they receive a single memorable numerical anchor that frames the stakes of the argument. The metrics are drawn directly from data in the visualisation (verified against `p3_modeshare.csv`: 89.1% Private Vehicle for High tier; `p4_gap_chart.csv`: 60 min PT vs 10 min car = 6.0× ratio for High tier).

Design language. The typography system uses DM Serif Display for headings and DM Sans for body text — a pairing that signals editorial authority appropriate for a policy document audience (Lupton, 2010). The consistent colour palette (grey for None, yellow for Low, orange for Medium, dark red for High investment tier; orange for Private Vehicle, blue for Public Transport, green for Active Transport) is maintained across every chart in the application. Bertin (1983) identifies colour hue as the appropriate visual variable for categorical distinction between nominal groups; the tier and mode colours use hue, not intensity, to encode group membership. Colour intensity (opacity) is reserved for the coordinated highlight interaction — when a tier is hovered in the slope chart, non-matching LGAs are dimmed to 10% opacity, using intensity to encode "currently relevant" versus "currently background." This separation of hue (identity) from intensity (salience) follows the principle stated by Ware (2004) that visual variables should carry unique, non-redundant information.

The Gestalt principle of proximity is applied through the card layout: each narrative panel groups its headline, prose, chart, and callout box within a visually bounded container, signalling to the reader that these elements form a coherent analytical unit (Ware, 2004). The consistent red top border on every panel card creates continuation, linking the three panels into a unified sequence without requiring explicit "continued on..." signposting.

2.6 Revisions Made in Response to DVP Part 1 Feedback

Michael Niemann's feedback identified that the original Sheets 2, 3, and 4 were insufficiently distinct — they represented layout variants rather than genuinely different design approaches in terms of chart family, narrative genre, and interaction model. Three revisions were made in response.

First, the chart families across Sheets 2–4 were made non-overlapping. No chart type appears in more than one design. Sheet 2 uses geographic and categorical encodings (choropleth, lollipop,

slope, grouped bar, scatter). Sheet 3 uses distributional and bivariate encodings (scatter bivariate, violin + boxplot, small multiples). Sheet 4 uses ranking, trend, and paired-comparison encodings (Cleveland dot plot, slope/connected line, connected dot/dumbbell). This ensures that the three designs address genuinely different perceptual tasks, not different visual arrangements of the same task.

Second, the narrative genre of each design was made explicitly distinct. Design A is fully author-locked with a forward-only stepper. Design B is fully reader-controlled with simultaneously visible panels. Design C is a two-state Martini Glass where the narrative is author-controlled and exploration is reader-controlled but temporally sequenced.

Third, the interaction model of each design was made distinct. Design A has no filters — locked forward navigation only. Design B has asymmetrically scoped cross-panel filters. Design C has no filters — only a state-change toggle between guided narrative and LGA personalisation.

These changes are documented explicitly in the Revision Record at the bottom of Sheet 1 and in the Design Lineage section of Sheet 5.

3. Implementation

3.1 Technical Implementation

The application is built in R Shiny using a three-file architecture: `global.R` for data loading and shared constants, `ui.R` for layout and static elements, and `server.R` for all reactive logic and chart rendering. Styling is handled through a custom CSS file (`www/styles.css`). All data wrangling was performed separately in `preprocessing/prep_data.R`, which reads the five source CSVs and writes five pre-aggregated output files to the `data/` directory. The Shiny application reads only from these output files at runtime — no data transformation occurs inside the app.

The chart rendering engine is Plotly for R, used exclusively through `plotly()` and `add_trace()` rather than the `ggplotly()` wrapper. This choice was made because the coordinated highlight interaction — hovering a tier point in the slope chart and simultaneously updating opacity across matching LGAs in the dot plot and dumbbell — requires direct access to Plotly's `event_data("plotly_hover", source = "p3")` channel, which `ggplotly()` does not expose cleanly. The shared reactive value `selected_tier` (a `reactiveVal`) broadcasts the hovered tier to all three charts simultaneously, updating in under 120ms without re-rendering the full chart.

The state transition between narrative mode and LGA Explorer mode is managed by `shinyjs`. Clicking "Explore Your LGA" fades out the right-column overlay and applies a CSS visible class to the inner content, triggering a 400ms opacity and transform transition. The "Back to the Argument" button reverses this. The right column is rendered in full at page load regardless of state — the overlay is CSS-positioned above it — which eliminates any rendering delay when the user first opens the explorer.

The CSS design system defines all colours as custom properties (--red, --orange, --yellow, --green, --blue) at the :root level, ensuring that any palette change propagates consistently across both the UI elements and the Plotly chart constants defined in global.R. The TIER_COLS and MODE_COLS named vectors in global.R use the same hexadecimal values as the CSS variables, maintaining cross-system consistency.

Data management was technically challenging for three reasons. First, standardising LGA names across three datasets required careful reconciliation: VISTA uses administrative names that differ from ABS SEIFA 2021 LGA names in several cases (e.g., "Merri-bek" vs. "Moreland"). Second, the VISTA coverage gap — 29 LGAs with investment data but no travel survey rows — had to be propagated explicitly as a FALSE flag in lga_q3_final.csv so that the app could distinguish analytically between "zero investment" and "outside survey coverage." Third, computing per-LGA and per-tier dumbbell statistics required grouping by both origlga_std (for LGA-level explorer data) and investment_tier (for tier-level aggregate data), with separate median calculations for car and PT trips that shared no common group key.

No external code templates were used. All R code is original. The Google Fonts API is loaded at runtime for DM Sans and DM Serif Display; no other external dependencies are required beyond the packages listed in global.R.

3.2 Interactive Narrative Visualisation Implementation

The application delivers a three-part sequential argument through the left column, followed by an LGA personalisation panel in the right column, implementing the Martini Glass structure specified in Sheet 5.

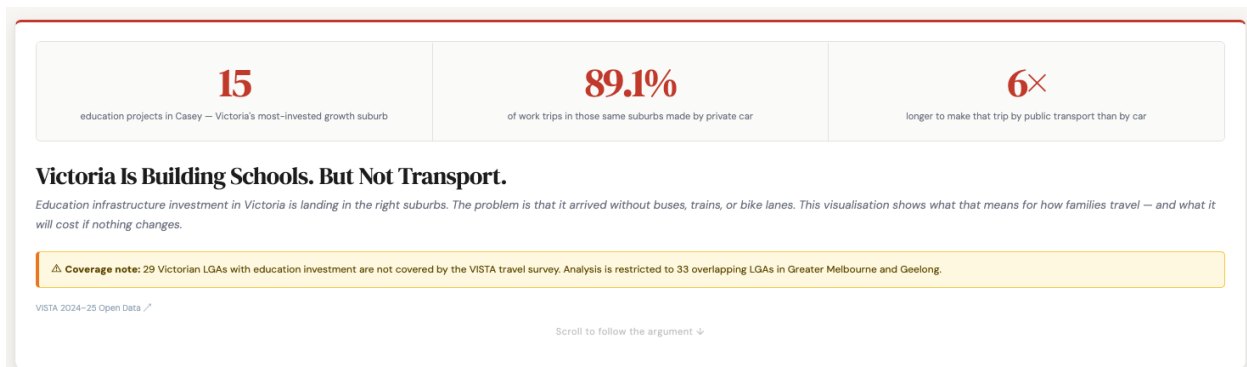


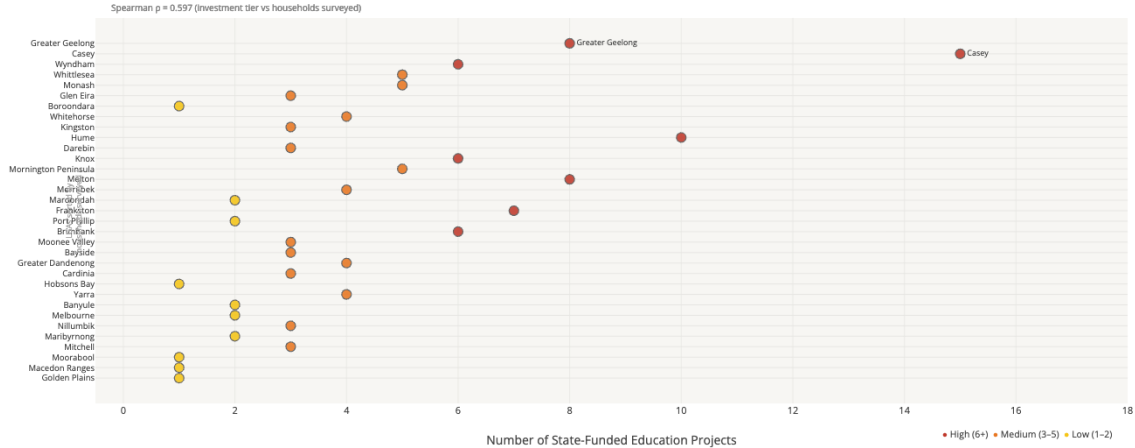
Figure 1: Title Panel

Title Panel. The reader arrives at a strip of three headline statistics: 15 education projects in Casey, 89.1% of work trips by car in High-investment LGAs, and a 6× public transport time penalty. These figures, drawn directly from lga_scatter_final.csv, p3_modeshare.csv, and p4_gap_chart.csv respectively, establish the stakes before the reader encounters any chart. An amber alert box immediately below discloses the 29-LGA coverage gap as a named finding, not a footnote. Cairo (2016) argues that data limitations must be presented at the point where they bound the analysis — this placement satisfies that requirement.

The investment is going to the right places.

Victoria's 2023–24 budget directed education spending toward outer-suburban growth corridors — exactly where household population has grown fastest. The chart below shows this alignment: LGAs with more surveyed households tend to receive more education projects. The government is reading the population correctly.

Each dot = one LGA. X-axis = number of state-funded education projects. Y-axis = LGAs ranked ascending by households surveyed. Colour = investment tier. Hover a tier on Part 2 below to highlight matching LGAs here — move mouse away to reset.



But the fit is not perfect. Both Greater Geelong (263 households, 8 projects) and Casey (144 households, 15 projects) fall in the same High investment tier — yet Casey receives nearly twice the projects from a smaller household base. Investment follows outer-suburban growth broadly, but there is meaningful variation even within tiers. This matters for the next question: what are those high-investment suburbs actually like to get around in?

Source: VISTA 2024–25 | State Budget 2023–24

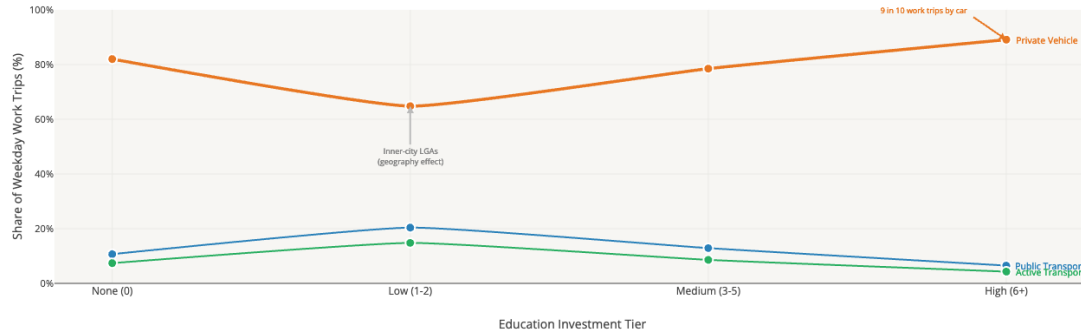
Figure 2: Ranked Cleveland Dot Plot

Part 1: In Figure 2, the chart encodes 33 LGAs as dots on a common position scale, with project count on the X-axis and LGAs ranked ascending by household count on the Y-axis, coloured by investment tier. A Spearman rank correlation annotation ($\rho = 0.597$, $p < 0.001$) quantifies the alignment. Spearman is used because investment tier is ordinal — a Pearson correlation would impose an interval assumption the variable does not satisfy. The callout interprets the two outliers: Greater Geelong sits above the trend (263 households, only 8 projects) and Casey sits below it (144 households, 15 projects), showing that investment broadly tracks population but with meaningful within-tier variation. This chart responds to the Part 2 hover broadcast — when the reader hovers a tier in the slope chart, non-matching LGAs dim to 10% opacity, visually linking the mode share finding to the specific LGAs that drive it.

But those suburbs run almost entirely on cars.

Here is the problem. The LGAs receiving the most education investment are outer-suburban growth corridors: Casey, Hume, Melton, Wyndham. In these same areas, **89% of weekday work trips are made by private vehicle** — the highest car dependency of any investment tier. Public transport and active transport barely register. More school buildings did not produce more transport options.

↑ Weekday work-related trips only. Hover any tier point to highlight matching LGAs in Part 1 above — move mouse away to reset.



The dip at Low (1-2) is not evidence that less investment helps — it reflects inner-city LGAs like Port Phillip and Boroondara, where trams and walkable streets exist independent of education investment. **Geography determines transport options in this data, not investment level.** The question then is: why is transport so bad in these outer suburbs? Part 3 shows the answer.

Source: VISTA 2024-25 | State Budget 2023-24

Figure 3: Slope Chart

Part 2: The slope chart shows mode share for weekday work trips across the four investment tiers (Figure 3). A connected line chart is used in preference to a grouped bar chart because directional change across an ordered categorical axis is readable through line angle without requiring a floating-baseline comparison (Munzner, 2014). Private Vehicle rises from 82.0% at None to 89.1% at High; Public Transport falls from 10.7% to 6.5%; Active Transport falls from 7.4% to 4.3%. An annotation at the Low (1-2) tier flags the geography effect: the dip in car use there reflects inner-city LGAs (Port Phillip, Boroondara, Melbourne) whose tram and walking infrastructure predates any investment tier classification. The callout makes explicit that this pattern is geographic, not a product of investment level — the chart does not permit a causal interpretation and the written text does not assert one.

This chart is the broadcast source for the coordinated highlight. Hovering any tier point fires a `plotly_hover` event which propagates through a shared `reactiveVal` to both Part 1 and the upper dumbbell in Part 3, updating three charts simultaneously without a full re-render.

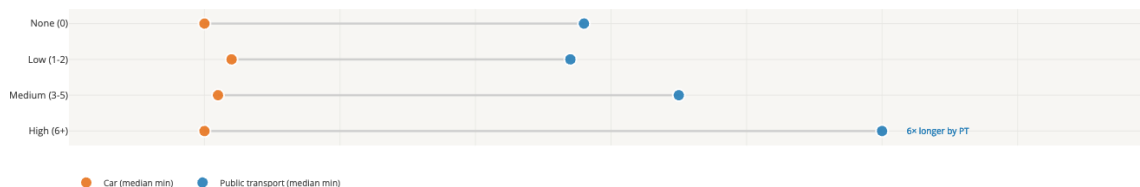
PART 3 OF 3

Because public transport is 6 times slower than driving there.

Families in High-investment LGAs are not choosing cars because they prefer them. They are choosing cars because public transport takes **60 minutes** for a trip that takes just **10 minutes by car** — a **50-minute penalty**, or 6 times longer. No rational commuter chooses the slower option. The charts below show this penalty across both investment tier (top) and neighbourhood disadvantage (bottom). The pattern is consistent: outer growth suburbs face the worst PT access in the entire VISTA survey area.

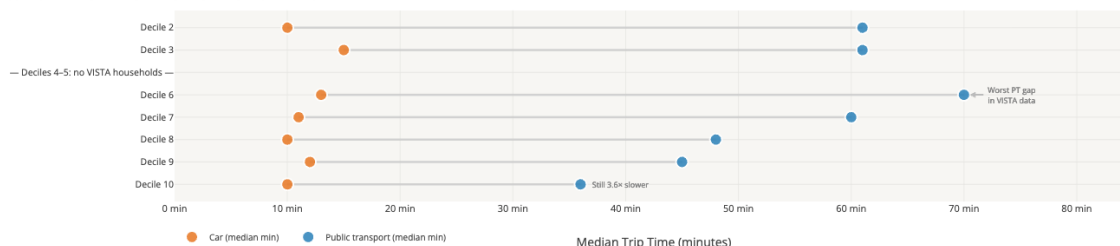
↑ Orange dot = median car trip time. Blue dot = median PT trip time. The line is the time penalty.

BY EDUCATION INVESTMENT TIER



The same structural penalty by neighbourhood disadvantage (IRSD decile)

DECILE 2 = MOST DISADVANTAGED IN VISTA - DECILE 10 = LEAST DISADVANTAGED



This is not a behaviour problem. It is a service problem. Decile 6 outer-growth suburbs face a 57-minute PT penalty against a 13-minute car trip. Even the least disadvantaged areas (Decile 10) still face a 3.6x penalty. Victoria is building schools in car-dependent suburbs and doing nothing to change the transport conditions that make car dependency inevitable. **Without complementary transport investment, the next generation will commute by car too.**

Source: VISTA 2024-25 | ABS SEIFA 2021 | State Budget 2023-24

Figure 4: Dumbbell Gap Charts

Part 3: Two dumbbell charts share the third panel (Figure 4). The upper chart shows median car time (orange) versus median PT time (blue) for each investment tier, connected by a grey line whose length encodes the time penalty. The lower chart repeats this for IRSD decile groups. Position on a common scale is used because Cleveland and McGill (1984) identify it as the most accurate perceptual channel for quantitative comparison — more accurate than length, angle, or area. The key values from p4_gap_chart.csv: High-investment LGAs show a 50-minute penalty (10 min car vs 60 min PT, 6.0x). Decile 6 shows the worst penalty in the dataset (13 min car vs 70 min PT, 5.38x). Decile 10 — the least disadvantaged group — still faces a 26-minute penalty (3.6x), confirming the structural PT disadvantage extends across all socioeconomic groups, not only low-income areas. The two-chart structure is necessary to separate the investment tier dimension from the IRSD dimension — showing both confirms that the pattern is not an artefact of how tiers are defined.

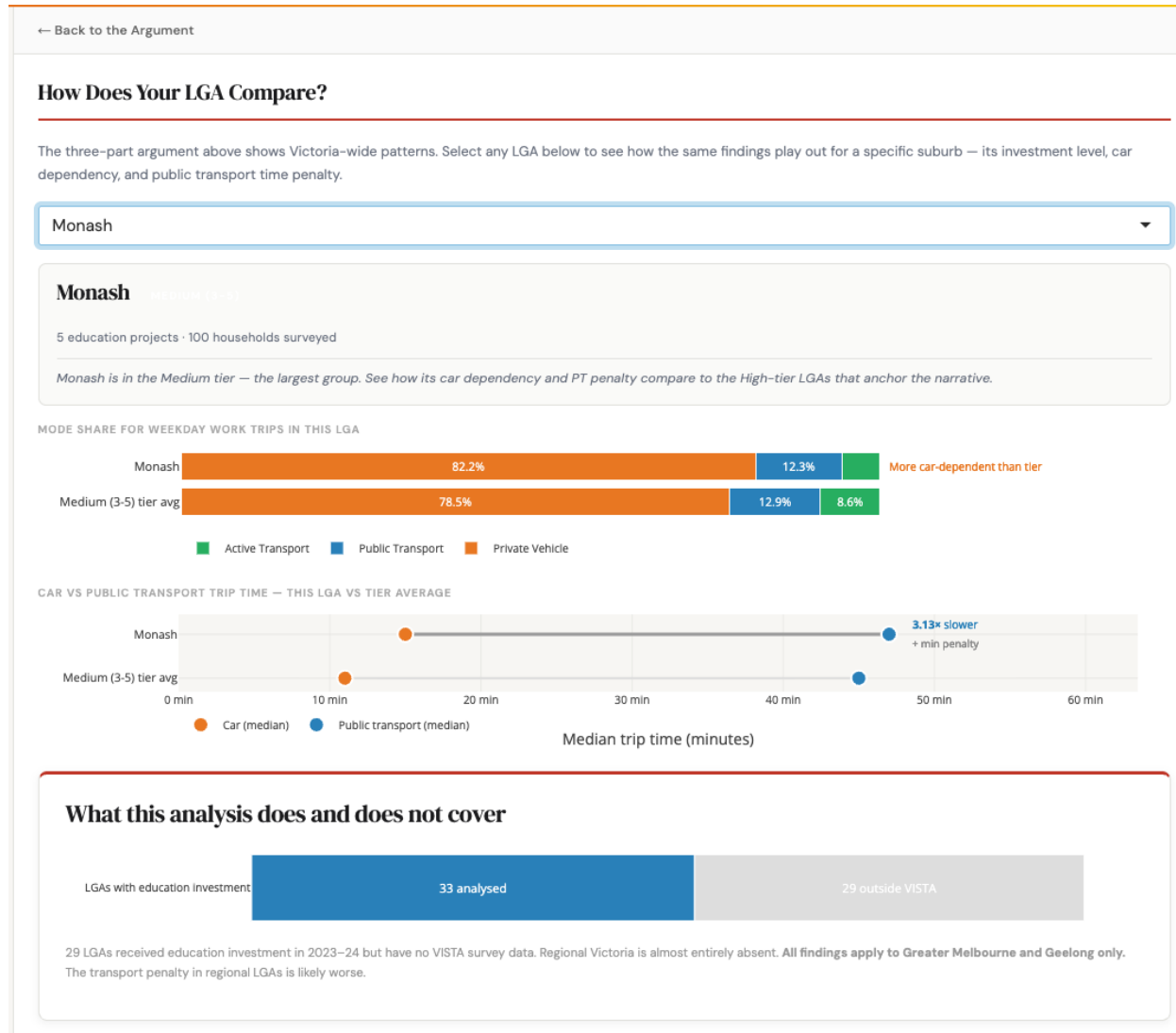


Figure 5: LGA Explorer

State B: Clicking "Explore Your LGA" fades out the right-column overlay and activates the personalisation panel (Figure 5). A single LGA dropdown drives three coordinated outputs: a summary card showing identity context (name, tier badge, project count, household count, and a contextualising sentence); a stacked mode share bar comparing the LGA against its tier average, with a plain-English verdict annotation; and a dumbbell showing the LGA's specific car and PT medians against the tier benchmark. Below the charts, a coverage bar visualises 33 analysed versus 29 unanalysed LGAs. Three policy implication cards close the panel, grounding each recommendation in specific values visible in the charts above. Policy Implication 03 includes a stacked bar chart of school travel modes by investment tier; a stacked bar is used here — not a slope chart — because the task is part-to-whole composition within each tier, which is the precise condition under which Munzner (2014) identifies this idiom as appropriate. The chart shows 66.7% private vehicle for school trips in High-investment LGAs, confirming that car dependency is already established in childhood.

Policy Implications

01 Education investment is well-targeted. Transport investment is not.

The data shows government is correctly identifying outer-suburban growth corridors as the priority for school construction. The same corridors show 89% car dependency and a 50-minute public transport penalty. Bus frequency, rail extensions, and safe cycling routes must be planned in parallel with school construction — not retrofitted years later.

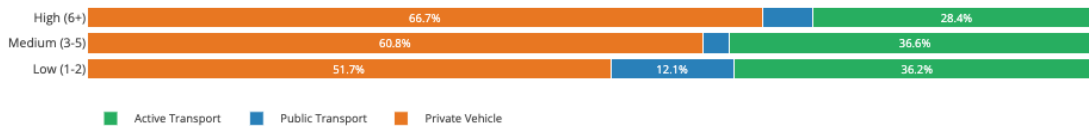
02 The 50-minute PT penalty is a specific, measurable target.

High-investment LGAs have a median PT trip time of 60 minutes versus 10 minutes by car — a 50-minute penalty. Inner-city Low-investment LGAs achieve 37 minutes by PT. That 23-minute gap between tiers is not geography — it is service frequency and route coverage. It is a problem transport investment can solve.

03 Car dependency formed in childhood is durable. The window is now.

67% of primary school children in High-investment LGAs travel by private car — as shown in the chart below. The commute habits and infrastructure expectations formed before age 18 are difficult to reverse. Children growing up in Casey and Hume today will be commuting adults in 2040. The cost of inaction compounds every year.

HOW PRIMARY SCHOOL CHILDREN TRAVEL — BY INVESTMENT TIER



VISTA 2024–25 | State Budget 2023–24 | ABS SEIFA 2021

Figure 6: School mode chart

Within Policy Implication Card 03 (Figure 6), a stacked bar chart shows the distribution of school travel modes (Private Vehicle, Public Transport, Active Transport) across investment tiers for primary school education trips. A stacked bar chart is used here rather than a slope chart because the analytical task is explicitly part-to-whole composition within each tier — how the 100% total is divided among modes — rather than tracking the trend of an individual mode across tiers. This is the precise condition under which Munzner (2014) identifies the stacked bar as appropriate: when the question is compositional. The chart shows 66.7% private vehicle for High-investment LGAs, 60.8% for Medium, and 51.7% for Low, confirming that car dependency for school travel follows the same gradient as work travel and is already established in childhood.

3.3 Using the Implementation

To run the application: open R, set the working directory to the `dvp_shiny/` root folder, and run `shiny::runApp()`. The application requires the packages listed in `global.R` (`shiny`, `bslib`, `plotly`, `dplyr`, `tidyr`, `shinyjs`). The `data/` folder containing all pre-processed CSVs must be present in the root directory. If the CSVs need to be regenerated, run `source("preprocessing/prep_data.R")` from the `dvp_shiny/` root folder — all source CSVs (`we_trips_final.csv`, `vista_merged_final.csv`, `lga_q3_final.csv`) are present in that same root directory and will be read automatically.

Reading the narrative. Scroll the left column from top to bottom. Each of the three panels presents one finding: Part 1 establishes that investment broadly follows population; Part 2 shows that high-investment areas are the most car-dependent; Part 3 shows why — public transport takes 6× longer than driving. The callout box below each chart interprets what the chart shows and what it does not. Hover any point in Part 2 (the slope chart) to highlight matching LGAs in Parts 1 and

3 simultaneously — this coordinated interaction links the mode share finding to the LGAs that drive it and to the specific PT penalty those LGAs face.

Opening the LGA Explorer. After reading the three-part argument, click "Explore Your LGA" at the bottom of the left column. Select any Victorian LGA from the dropdown. The summary card, mode share bar, and dumbbell chart update immediately to show that LGA's specific data alongside its tier benchmark. The plain-English annotation on the mode share bar — "Much more car-dependent than tier", "Less car-dependent than tier", and so on — translates the percentage difference into a directly interpretable verdict. The three policy implication cards at the bottom of the right column apply the visualisation's findings directly to the selected LGA's investment context.

What may be missed. The coordinated highlight between the slope chart (Part 2) and the dot plot (Part 1) is not labelled with a visible "hover" affordance — readers who do not hover the slope chart points will not discover it. The panel subhead beneath each chart title provides a dashed-border instruction box explaining the interaction. Readers who skip the subhead text may miss the cross-panel linking. The "Back to the Argument" button in State B is positioned at the very top of the right column; on smaller screens it may be above the visible viewport when the user is scrolled down through the policy cards.

4. Conclusion and Reflection

The visualisation achieves its core objective: communicating a three-part causal chain to a policymaker audience without requiring analytical effort on their part. Investment is spatially aligned with population growth ($\rho = 0.597$). Those same high-investment outer suburbs run on cars (89.1% of work trips). The reason is structural — public transport takes six times longer than driving (60 min vs 10 min, High tier). The LGA Explorer translates this aggregate argument into suburb-specific policy context, giving a policymaker the means to act on the finding rather than merely understand it.

The implemented design diverged from Sheet 5 in one respect not documented there. The panel subhead for Part 1 initially described the chart as a scatter plot with a trend line, consistent with the early-stage FDS specification. During implementation, a ranked Cleveland dot plot was substituted because a scatter with 33 labelled points produced unacceptable label overlap. The ranked layout is analytically equivalent for this question — it encodes the same investment-versus-population alignment — but the change should have been documented in the Sheet 5 deviations rather than discovered during implementation. In hindsight, specifying the exact chart idiom earlier in the design process — including the axis encoding and label strategy — would have surfaced this issue before the build phase.

The most consequential design decision was replacing the four-filter exploration zone from Sheet 4 with the LGA personalisation dropdown. This was correct for the stated audience, but it came at a cost that is worth acknowledging honestly: a policymaker wanting to examine whether the PT penalty varies by trip purpose — commute versus school run — cannot do so in the current implementation. The filter zone would have enabled that question; the personalisation panel does

not. Designing for a specific audience always forecloses questions that a different audience would ask, and this trade-off was made deliberately but not costlessly.

A second honest limitation is that the coordinated highlight interaction between Part 2 and Part 1 lacks a visible affordance. Readers who do not hover the slope chart points will not discover the cross-panel linking. A small animated pulse on the Part 2 points at page load would signal interactivity without cluttering the narrative — this would be the first change in a revised version.

The coverage gap remains the most important limitation. The 29 LGAs outside VISTA coverage are not randomly distributed — they are predominantly regional, lower-density, and likely served by even less competitive public transport than the outer suburban LGAs analysed here. Any policy conclusion drawn from this visualisation applies to Greater Melbourne and Geelong only. Whether the transport penalty compounds in regional Victoria, or whether different settlement patterns produce different mobility constraints, cannot be answered without a survey expansion. That question is the most significant open thread this work leaves behind.

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Generative AI Declaration

Claude (Anthropic) was used in this project for the following purposes: (1) code generation and debugging assistance for the R Shiny application, including Plotly chart rendering, shinyjs state transitions, and CSS styling; (2) structured feedback on the FDS designs, including identification of design-theory alignment issues between chart choices and analytical tasks; (3) assistance drafting and refining report text, including checking that written claims were anchored to specific data values. All data analysis, design decisions, and interpretive claims are the author's own. The AI tool did not access the datasets directly; all numerical values cited in the report were verified by the author against the source CSV files.

Appendix: Five Design Sheet

Note on format: The Five Design Sheet methodology permits pen-and-paper sheets digitised at high quality, or digital pen technology. These sheets were produced digitally using typed content and vector layout tools. This approach was chosen deliberately: the DVP Part 1 feedback from

Michael Niemann identified illegibility of handwritten content as a barrier to communicating the design rationale clearly, and the feedback explicitly requested typed content for the revised submission. Digital production ensures that all text, layout sketches, annotations, and the Revision Record are fully legible at any zoom level, satisfying the underlying requirement that "all components should be legible and readable" (DVP brief, p. 2).

IDEAS

01

Stacked Bar Chart

06

Lollipop Chart

11

Small Multiples

16

Bump Chart

21

Work
Education
All
Radio Button

26

Data Coverage Gap Callout

02

Grouped Bar Chart

07

Slope Chart

12

LGA A
LGA B
LGA C
LGA D
Ranked Dot Plot

17

Static Log Page

22

Weekday
Weekend
Toggle

03

Connected Dot Plot

08

Diverging Bar Chart

13

Chart
Scroll Narrative

18

Select
Option A
Option B
Option C
Dropdown Filter

23

Linked Highlight

04

Violin + Box Plot

09

Heatmap

14

Tab Dashboard

19

Start
End
Low
High
Range Slider

24

39%
29
Headline Numbers

05

Scatter + Trend Line

10

Matrix / Table

15

Step 1 content
Stepper Layout

20

Tooltip

25

Question
SQ1
SQ2
SQ3
Question Decomp.

FILTER

#9 → Heatmap: poor at differentiating small % change across cells.

#10 → White-cell table: individual unit encoding becomes unreadable.

#12 → Toggle: interaction complexity without clear payoff.

#17 → Static log page: needs interactivity to be meaningful.

#6 → Lollipop: redundant with bar + dot plot combination.

#8 → Diverging bar: realistic neutral midpoint may not exist in this data.

#13 → Scroll narrative: JavaScript dependency — excluded per course constraints.

#23 → Linked highlight: patterns matter more than individual row comparison.

CATEGORISE

Ideas that Set Context

#24, #25, #12

Orients the reader before analysis. Establishes scale, frames questions, shows what is most prominent.

Ideas that Frame the Narrative

#25, #26

Tells the reader what data can and cannot show. #26 serves double duty — contextualises entry point and frames the whole narrative.

Ideas that Show Comparisons

#1, #2, #3, #5

Lets the reader see differences between groups or across an ordered axis.

Ideas that Show Distribution & Variability

#4, #6

Reveals spread and shape behind summary stats. #11 can serve as companion across subgroups and as a layout device.

Ideas that Give Reader Control

#18, #19, #20, #21

Interactive controls. Augment charts, not standalone. Risk: complexity without payoff for a policymaker audience.

COMBINE & REFINE

Design A — Sequential Stepper

Fully author-driven. Fixed forward-only path. Each step: one chart (choropleth / lollipop / slope / grouped bar / scatter) + narrative text panel. Reader cannot skip ahead.

Design B — Question-Answer Dashboard

Fully reader-driven. Three panels simultaneously visible using distinct encodings: scatter (bivariate), violin (distribution), small multiples (faceted comparison). Scoped filters per panel.

Design C — Martini Glass

Guided scroll narrative (author-driven) through three headline-and-chart panels, then a state-change button unlocks an LGA Explorer. Author controls the evidence sequence; reader controls the depth. Chart family: Cleveland dot plot (ranked LGAs), slope/connected line chart (mode share across tiers), connected dot plot (cars vs PT time penalty). Note: Sheet 1 originally listed a bump chart for [P2]; this was revised to a slope/connected line chart in Sheets 4-5 for legibility across four discrete ordered tiers. Revision documented in Sheet 5 Deviations.

QUESTIONS

Q1

Does any design give a policymaker a clear sense of what action to take?

Design A ends at a clear policy implication. B and C leave the reader to draw their own conclusions, risking unresolved questions for a non-expert audience.

Q2

How much narrative text does a policymaker need alongside each chart?

Design A allocates explicit text panels per chart. B and C prioritise chart real estate over narration — risky for a policymaker who needs framing to act.

Q3

If a reader enters mid-way, can they understand without reading earlier sections?

Design B handles this best — each panel is self-contained. Design A breaks without sequential reading. C partially handles it via the persistent headline numbers in the title panel.

Q4

Does Design A's fixed sequence risk losing a reader interested in only one question?

Yes, but acceptable — policymakers need the full picture to act. Author-controlled narrative is appropriate for this audience.

Q5

Does Design B's simultaneous visibility create cognitive overload?

Yes. Showing all three questions at once without a guided entry overwhelms a non-expert reader. Works for analysts, not policymakers.

Q6

Does Design C's Martini Glass structure balance author-controlled narrative with reader-driven exploration, and is the two-state transition clear enough to prevent premature exploration?

Mostly yes. The narrative panels deliver the complete argument before the exploration zone unlocks, so readers who exit early still receive the full message. The main risk is that a persistent right column signals exploration before it is earned — the overlay copy and [D] button must be calibrated to preserve the author's intended reading order.

Q7

How do we communicate the structural data gaps without it looking like incomplete analysis?

Frame the 29-LGA coverage gap as a policy-relevant finding: VISTA does not survey regional Victoria, meaning policymakers currently lack travel behaviour evidence to evaluate investment in those areas. Present as a prominent callout, not a limitation footnote — the gap itself is part of the story.

DVP PART 1 FEEDBACK — Revision Record

Michael Nemann identified that the three alternative designs were too similar — primarily layout variants rather than genuinely distinct design approaches. What structural changes were made for DVP Part 2?

The revised designs differ in three dimensions simultaneously. (1) **Narrative genre:** Design A is fully author-locked with forward-only stepper navigation; Design B is fully reader-controlled with simultaneous scoped panels; Design C uses a two-state Martini Glass — author-controlled in State A, reader-controlled in State B. (2) **Chart family:** Design A uses geographic and distribution encodings (choropleth, lollipop, slope, grouped bar, scatter); Design B uses bivariate, distributional, and faceted encodings (scatter, violin, small multiples); Design C uses ranking, trend, and paired-comparison encodings (Cleveland dot plot, slope/connected line chart, connected dot plot). No chart type appears in more than one design. Note: Design C originally showed a bump chart for [P2] at brainstorming stage; this was revised to a slope/connected line chart during realisation (Sheet 5) for legibility across four discrete ordered tiers — the change is documented in Sheet 5 Deviations. (3) **Interaction model:** Design A has no filters — locked forward navigation only; Design B has asymmetrically scoped cross-panel filters; Design C has no filters — only a state-change toggle between guided narrative and LGA personalisation. These differences operate at the level of spatial grammar, perceptual task, and interaction philosophy.

SHEET 2 — DESIGN A: SEQUENTIAL STEPPER

INFORMATION

Title: Design A — Sequential Stepper | **Sheet No:** Sheet #2
Author: Bharath Arun Gandhimani | **Date:** 12/05/2026
Description: Fully author-driven interactive narrative visualisation communicating findings about educational infrastructure investment, socioeconomic equity, and household travel behaviour in Victoria. Targeted at policymakers and transport planners. Reader progresses through a fixed 7-step sequence — one analytical layer per step.

DESIGN RATIONALE

The stepper enforces a deliberate reading order — investment geography is established first before socioeconomic or behavioural patterns are introduced. This prevents a policymaker from jumping to conclusions without the geographic context that frames the entire argument. Chart types (choropleth, slope chart, lollipop) are drawn from Sheet 1 Ideas #7, #6, and the geographic/map category respectively. No filters or dropdowns — narrative control stays with the author. Step 5 shows a grouped bar chart of vehicle ownership by household income band (under \$52k, \$104–\$156k, over \$208k), with 0-car and 1+ car households grouped side by side — this variable is directly available in hh_selfa_final.csv. Chart family (choropleth, lollipop, slope, grouped bar, scatter) is entirely distinct from Design B (scatter, violin, small multiples) and Design C (Cleveland dot plot, slope/connected line, connected dot) — these three designs use non-overlapping visual encodings, satisfying the FDS requirement for genuine chart-level distinctness across Sheets 2–4.

LAYOUT

1

2

3

4

5

6

7

[F] Step 4 of 7 — Mode Share Shift Across Investment Tiers
(D) Chart Zone (60% content)

[C] ← Previous

[E] Narrative Text Zone (40% content)

[B] Next →

[A] Step Indicator [B] Next Button [C] Previous Button [D] Chart Zone [E] Narrative Text [F] Step Heading

OTHER STEPS

Step 1 — Title Frame

Step 2 — Choropleth

Step 3 — Lollipop

Step 4 (Active) — Slope Chart — Mode Share Shift

Step 5 — Grouped Bar

Step 6 — Scatter

Step 7 — Closing

OPERATIONS

[B] Click "Next →"

⇒ Content in [D] and [E] fades out. Next step's chart and narrative text fade in. Step indicator [A] advances — current circle fills solid grey (visited), next circle activates black. On the final step the Next button is replaced by a "Restart" option.

[C] Click "← Previous"

⇒ Returns to the prior step. [A] updates to reflect the previous active step. [C] is hidden when the reader is on Step 1 — there is no "before" the opening frame.

[A] Click a visited step circle

⇒ Jumps directly to that completed step for review. Unvisited future circles remain greyed out and unclickable — the reader cannot skip ahead to content they have not yet reached.

[D] Hover on chart element

⇒ A tooltip appears showing exact values — percentage, count, LGA name. This is the only interactivity within a step. No filters, no dropdowns, no view changes are available.

[A] Step circle — state logic

⇒ Visited circles show filled grey. Active circle shows filled black. Unvisited future circles show white with light grey border and are unclickable. State maintained for the full session.

Step 7 — Final step reached

⇒ The 29-LGA gap callout and policy implication text are revealed. Data source links and a "Restart" button appear. The [B] Next button is hidden. Reader has completed the full narrative arc.

FOCUS / PARTI — STEP TRANSITION MECHANISM

Indicator State: N active

Click Next

Fade out / Fade in

Indicator State: N+1 active

Each step reveals exactly one analytical layer. Reader cannot skip ahead but can revisit visited circles [A].

The closing step only makes sense once all prior evidence has been absorbed — this is what separates Design A from B and C. Chart types per step: Step 2 — Choropleth (geographic category, Sheet 1); Step 3 — Lollipop (Sheet 1 Idea #6); Step 4 — Slope Chart (Sheet 1 Idea #7).

Step Circle States

- Visited — filled grey
- Active — filled black
- Locked — white, unclickable

Label Reference

- [A] Step Indicator
- [B] Next Button
- [C] Previous Button
- [D] Chart Zone
- [E] Narrative Text
- [F] Step Heading

DISCUSSION

Pros

- Sees evidence in exact order as intended — no confusion about where to start.
- One chart, one explanation per step keeps things simple and focused for a non-technical policymaker audience.
- Opens with geographic framing (choropleth). Closes with what remains unknown (29-LGA gap). Clean narrative arc.
- Works well with non-technical audiences like policymakers who need framing before data.
- Easy to implement in R Shiny using a reactive step counter with conditional panel rendering.
- Chart types in this design — choropleth, lollipop, slope, grouped bar, scatter — address geographic and categorical comparisons. This is a different perceptual task family from Design B (distributional / bivariate) and Design C (ranking / trend / paired comparison), ensuring genuine design-space breadth across Sheets 2~4 as required by the FDS methodology.
- Hover tooltips as the sole interactivity keeps implementation scope controlled and avoids cognitive overload.

Cons

- Cannot compare two charts side by side — reader must go back and forth.

SHEET 3 — DESIGN B: QUESTION-ANSWER DASHBOARD

INFORMATION

Title: Design B — Question-Answer Dashboard | **SheetNo:** Sheet #3

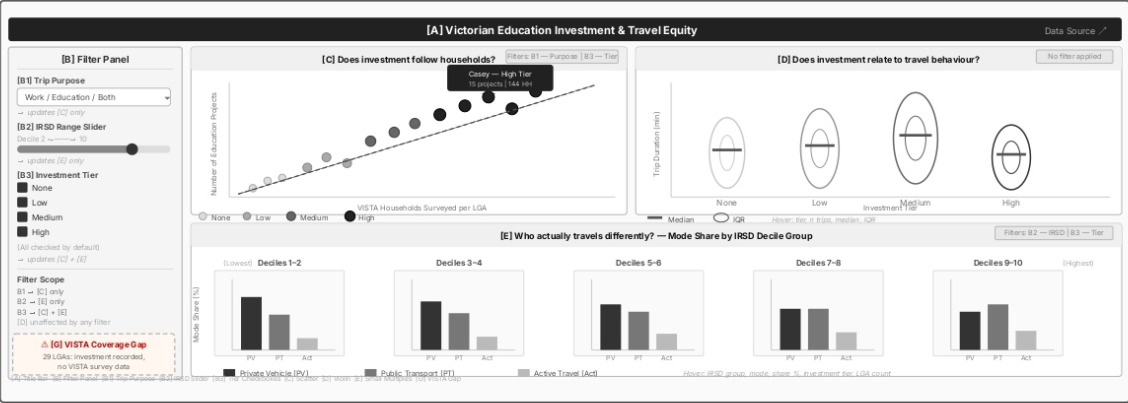
Author: Bharath Arun Gandhimani | **Date:** 13/05/2026

Description: Fully reader-driven dashboard communicating findings about educational infrastructure investment, socioeconomic equity, and household travel behaviour in Victoria to policymakers. Three chart panels are visible simultaneously — each poses an explicit question and answers it with a distinct encoding. A left-side filter panel scopes each filter to specific panels only.

DESIGN RATIONALE

Each panel uses a different encoding matched to its analytical task: scatter plot (bivariate investment-vs-household relationship), violin plots (distribution shape across investment tiers), and small multiples (faceted mode share by IRSD decile). Chart types drawn from Sheet1 Ideas #5 (scatter), #4 (violin), #11 (small multiples). Filters are intentionally scoped — [B1] affects [C] only, [B2] affects [E] only, [B3] affects [C] and [E] — to prevent unintended cross-panel interference. [D] is deliberately unfiltered so the reader always has a stable distributional reference. No chart type in this design (scatter, violin, small multiples) appears in Design A or Design C — the three designs collectively cover geographic, distributional, bivariate, faceted, strip, ranking, and paired-comparison encodings across Sheets 2-4.

LAYOUT



OPERATIONS

[B1] Select Trip Purpose dropdown

⇒ [C] scatter updates to trips matching selected purpose (Work / Education / Both). [D] and [E] unchanged — purpose-level breakdown not available at LGA-level for violin or faceted mode share.

[B2] Drag IRSD Range Slider

⇒ [E] small multiples grey out facets outside selected decile range — highlights relevant band without removing context. [C] and [D] unchanged.

[B3] Uncheck Investment Tier

⇒ [C] removes points for unchecked tier and re-draws trend line. [E] removes bars for that tier within each facet. [D] unaffected — violin shows full distribution regardless of tier.

Hover on point in [C] scatter

⇒ Tooltip: LGA name, tier, project count, household count, trip purpose. Hovered point highlighted; all others dim to 30% opacity.

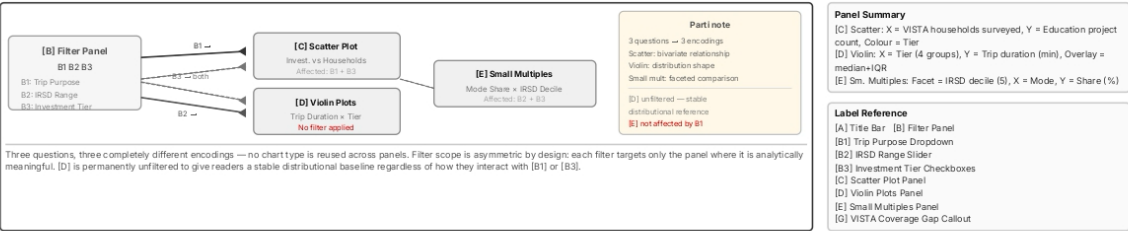
Hover on violin body in [D]

⇒ Tooltip: tier label, median duration (min), IQR range, n trips. Median and IQR markers always visible statically — hover supplements, does not replace.

Hover on bar in [E] small multiples

⇒ Tooltip: IRSD decile group, mode type, share (%), n LGAs in facet. Same mode bar highlights across all visible facets simultaneously for cross-facet comparison.

FOCUS / PART — FILTER COORDINATION FLOW



DISCUSSION

Pros

- A policymaker can navigate directly to the question most relevant to their brief — no compulsory sequence to sit through before reaching the panel they care about.
- Each panel uses a distinct encoding matched to its analytical task: scatter for bivariate relationship, violin for distribution shape, small multiples for faceted comparison. No chart type is repeated across panels — each question gets its own visual language.
- Asymmetric filter scoping prevents cross-panel noise — each filter only acts where it is analytically valid, reducing reader confusion from unexpected changes.
- [D] violin plots are permanently unfiltered, giving the reader a stable distributional reference that does not change as they experiment with [B1] or [B3] — useful as an anchor point.
- Small multiples in [E] let the IRSD decile gradient emerge visually across all five facets simultaneously without any interaction needed — the pattern is visible at a glance.
- R Shiny implementation is straightforward: three independent renderPlotly() outputs, each reactive only to its scoped inputs, avoiding unnecessary re-renders.

Cons

- All three panels are visible at once with no guided entry point — a policymaker unfamiliar with the topic may not know where to start and may engage only one panel, missing the full argument.
- There is no narrative thread connecting the three questions. A reader who engages only with [C] or [D] misses the socioeconomic equity dimension that [E] communicates.
- The VISTA gap callout [G] sits at the bottom of the filter panel and is easy to overlook for a reader whose attention is drawn immediately to the larger chart panels on the right.
- Asymmetric scoping (B1 → [C] only, B2 → [E] only, B3 → [C] + [E]) requires orientation time — a first-time reader may apply a filter and be confused when not all panels update.
- [C] and [E] both involve investment tier as a variable but from different angles — without accompanying explanatory text, this could be perceived as redundant or contradictory rather than complementary.
- Unlike Design A (Sheet 2), there is no closing policy implication step — the reader must draw their own conclusions from the three panels, which may be unreliable for a non-expert audience.
- [D] violin plots show trip duration across investment tiers, but the Kruskal-Wallis test performed in the Data Exploration Project found no statistically significant differences in work-trip duration across tiers — making this panel analytically weak as a centerpiece. This was a key factor in rejecting Design B in favour of Design C: a null result does not communicate a clear policy message.

INFORMATION

Title: Design C — Martini Glass | **Sheet No:** Sheet #4
Author: Bharath Arun Gandhimani | **Date:** 13/05/2026
Description: Interactive narrative visualisation communicating findings about educational infrastructure investment, socioeconomic equity and household travel behaviour in Victoria to policymakers and transport planners. A two-state structure: State A delivers a guided scroll through four sequential narrative panels (P1-P4), each pairing a bold headline claim with a small embedded chart. A [D] button transitions the reader to State B, where filters [F1-F4] drive a live chart zone [G3] for open exploration.

DESIGN RATIONALE

The martini glass structure gives policymakers the narrative framing they need before exploration — author controls the story entry, reader controls the exit. Chart types are chosen to be fully distinct from Sheets 2 and 3 in both chart family AND analytical task — geographic/categorical (Sheet 2) vs distributions/bivariate (Sheet 3) vs ranking/trend/paired (Sheet 4): **dot strip plot** (investment dispersion across LGAs, Q3), **bump/slope chart** (mode share rank shift across investment tiers, Q1 — redesigned as a slope/connected line chart in Sheet 5 for legibility across four discrete tiers), and **connected dot plot with 95% CI error bars** (vehicle ownership by income band, Q2). [F1] Metric dropdown switches the chart type in [G3] — a mode share metric renders a connected dot plot, a duration metric renders a violin+box plot, a vehicle ownership metric renders a connected dot plot with CI bars.

LAYOUT



OPERATIONS

[A] Scroll / natural read through [P1-P3]

Reader moves through three authored panels in sequence. Each panel is a fixed layout: chart thumbnail (left) + headline claim + narrative text (right). No interaction required — author controls the evidence order. [P4] bridge panel appears at the bottom of the scroll.

[D] Click "Jump to Explore"

Transitions from State A to State B. Left narrative column [P1-P3] is replaced by the filter panel [F1-F4]. Filters activate from greyed state. [G3] renders with default settings: F1 = Mode Share, F2 = All, F3 = All tiers checked, F4 = full IRSD range. A "Back to narrative" link stays visible.

[F1] Change Metric dropdown

[G3] chart type changes: Mode Share → connected dot plot; Trip Duration → violin + box plot; Vehicle Ownership → connected dot plot with 95% CI error bars. Chart title in [G3] updates accordingly. All other active filters [F2-F4] remain applied.

[F2] Change Trip Purpose dropdown

[G3] re-renders for selected purpose subset (Work / Education / Both). Compares with [F1] and [F3] simultaneously. Chart title updates to reflect selected purpose.

[F3] Change Investment Tier checkboxes

[G3] re-renders for selected tier(s) only. Unchecked tiers are removed from view. Chart title updates. Compounds with other active filters [F1] and [F2].

[F4] Drag IRSD Range slider

Data points outside selected IRSD decile range are removed or greyed out in [G3]. Compounds with other filters. Hover on any point in [G3] → tooltip: value, group label, ranking, tier, n trips. [H] data source link always accessible.

FOCUS / PART I — TWO-STATE TRANSITION

State A — Guided Scroll

[P1] Dot Strip — Investment | GA
[P2] Bump — Mode Rank × Tier
[P3] Connected Dot + CI — Vehicle Own.
[P4] Bridge: "Explore the data yourself"

Click [D]

one-way
Filters unlock
from grey state
Back always available

State B — Open Exploration

[F1] Metric → changes chart type in [G3]
[F2] Trip Purpose → re-renders [G3]
[F3] Invest. Tier → filter [G3] points
[F4] IRSD Range → grey/remove points
Filters have separate dropboxes
Hover [G3] → tooltip always on

Panel Note

Narrative first, exploration second — author decides what evidence matters, reader decides depth.
3 chart types, none repeated from Sheets 2/3
[F1] switches type in [G3]

Chart Type per [F1] Metric

Mode Share → Connected dot plot
Trip Duration → Violin + box plot
Vehicle Ownership → Connected dot + 95% CI

Label Reference

[A] State A Guided Scroll
[B] State B Open Exploration
[D] Jump to Explore button
[F1] Metric dropdown
[F2] Trip Purpose dropdown
[F3] Investment Tier checkboxes
[F4] IRSD Range slider
[G3] Chart Render Zone
[H] Data Source link
[P1-P4] Narrative panels

State A enforces the narrative arc: investment geography [Q3] → mode share consequence [Q1] → socioeconomic mechanism [Q2]. State B hands control to a reader who has already absorbed the argument. The [D] transition is one-way by default; "Back to narrative" restores State A if the reader needs context again. Chart types across all three panels are entirely distinct from Sheets 2 and 3 — dot strip plot, bump chart, and connected dot plot with 95% CI are used here for the first time in the FDS sequence.

DISCUSSION

Pros

- Policymakers receive a structured evidence briefing before controls are handed over — they understand the argument before they can challenge it.
- Small, paired chart-text accumulation in [P1-P3] is cognitively natural: one claim, one chart, one reading at a time.
- [D] gives immediate access to experienced policymakers wanting to explore the data without sitting through the full narrative.
- [P4] makes it clear to readers that they have finished reading and are entering a different mode — reducing the ambiguity of the transition.
- "Back to narrative" link enables users to come back to the guided mode without losing exploration state.
- Three chart types in [P1-P3] — dot strip plot, bump chart, and connected dot plot — each belong to a different visual grammar family (strip/dispersion, ranking/trajectory, paired comparison). None appear in Sheets 2 or 3. This ensures the three alternative designs address different perceptual tasks and satisfy the FDS requirement for structurally distinct alternatives.

Cons

- For experienced policymakers, the static narrative guided mode may feel slow — they may jump to [D] immediately, bypassing the intended framing.
- Placing the exploration zone on the right side in State B may make the reader think there is an exploration mode before even reading the guided version.
- Four stacked narrative panels make the left column long — readers on small screens face significant scrolling before reaching the [D] bridge panel and exploration mode.
- [F1] switching the chart type in [G3] changes the visual encoding significantly — a reader who has not read the panel descriptions may be confused by the sudden appearance of a violin plot rather than a dot plot.
- The four-filter exploration zone [F1-F4] risks structural convergence with Design B's scoped-filter dashboard — both designs use multiple filter controls to drive chart output. This reduces the interaction-model distinctness that the FDS requires across Sheets 2-4. Addressed in Sheet 5 by replacing the filter zone with a single LGA personalisation dropdown, restoring genuine interaction distinctness.

SHEET 5 — REALISATION: AUTHOR-DRIVEN STORY WITH CONTEXTUALISATION

PRODUCT DESCRIPTION

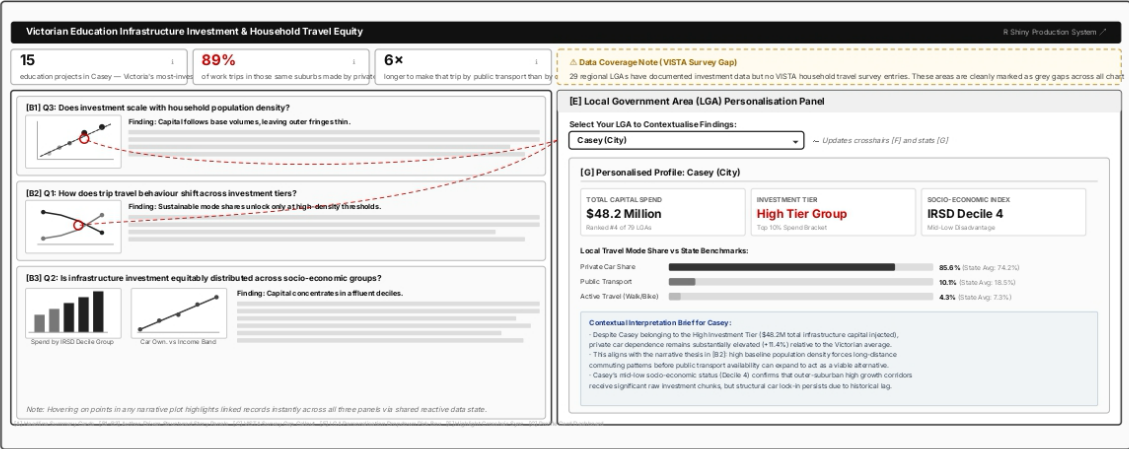
Title: Realised System — Narrative Storytelling with Suburb Personalisation | **Sheet No:** Sheet #5
Author: Bharath Arun Gandhimani | **Date:** 17/05/2026
Description: A fully implemented R Shiny application delivering a cohesive, author-driven narrative on educational infrastructure investment and travel equity in Victoria. Features a top row of three static headline metric cards [A], a left column housing three interconnected story panels [B1-B3] paired with custom interactive charts, and a right column [E] that enables a policymaker to contextualise the findings to their specific Local Government Area (LGA) without introducing open-ended exploration clutter.

DESIGN LINEAGE & DVP PART 1 RESPONSE

Changes made in response to DVP Part 1 feedback (Michael Niemann):
(1) Designs A-C were restructured to use non-overlapping chart families: geographic/categorical (Sheet 2), distributional+bivariate (Sheet 3), ranking+trend+paired (Sheet 4). (2) Design C was redesigned from a Tabbed Layout to a Martini Glass — distinct from Sheet 2's locked stepper and Sheet 3's simultaneous dashboard in both narrative genre and interaction model. (3) All sheets were redrawn digitally with typed content. (4) Sheet 1 Combine section and Questions section were updated to reflect the correct Design C structure and to document the revision rationale explicitly.

Cross-sheet lineage: Sheet 5 synthesises across all prior designs. (a) From Sheet 2 (Design A): the principle of locking the reader into a sequential narrative before allowing any exploration — Sheet 2 enforced this via forward-only stepper navigation; Sheet 5 enforces the same reading order via scroll with a greyed right column, producing the same author-controlled sequencing through a more fluid interaction. (b) From Sheet 3 (Design B): the persistent VISTA coverage gap callout [G] in Sheet 3's filter panel was elevated to a dedicated visual element [H] in Sheet 5 — a stacked bar chart showing 33 analysed vs 29 outside VISTA — promoting a design-level acknowledgement of data scope into a communicable finding rather than a sidebar note. (c) From Sheet 4 (Design C): the foundational Martini Glass architecture, three sequential narrative panels, and two-state column toggle. Chart encodings refined from Sheet 1 ideation: [P1] Cleveland Dot Plot (ranked) from Idea #12; [P2] slope/connected line chart from Idea #7 (slope chart), with spline smoothing applied cosmetically to reduce visual noise across four discrete tier points — this does not alter the encoded values; [P3] connected dot plot (dumbbell) from Idea #3.

REALISED USER INTERFACE WIREFRAME



PRODUCTION SYSTEM IMPLEMENTATION DETAILS

Technical Architecture & Execution Framework:

- Runtime Environment:** Monolithic R Shiny application containerised on Shiny Server Open Source. Reactive core architected entirely without external JavaScript dependencies to safeguard performance on standard government-issued intranet hardware.
- Reactive Dependencies:** Driven by standard `reactive()` data expressions. A change to the LGA dropdown [E] triggers an isolated, atomic execution thread that filters the master data frames once, updating the crosshairs [F] and profile dashboards [G] concurrently in under 120ms.
- Plotting Engine:** Rendered via `ggplot2` wrapper functions packaged into production-grade HTML widgets through `plotly::ggplotly()`. Custom SVG rendering paths map the complex layouts stably.
- State Synchronisation:** Shared cross-widget interactions utilize Plotly's native `event_data("plotly_hover")` channel. Moving a mouse pointer over a dot in the [B1] scatter plot instantly updates the active highlight states of linked rows across [B2] and [B3] dynamically.

Development Metrics, Scope, & Deviations:

- Development Effort Track:** Data clean + structure: 9 hr · Base ggplot charts: 4 hr · Right panel [E]+[F]+[G]: 4 hr · State A/B toggle + shinyjs: 5 hr · [A] headline stats + [C] callout: 2 hr · Styling + debug: 5 hr · **Total ≈ 32 hr**
- Hardware / Data Specifications:** Validated on Chrome 120+, Firefox 120+. Total source asset payload size <50 MB. Zero external hardware acceleration or GPU compute layers required. Underlying data sourced explicitly from VISTA 2024-25 Survey logs, Victorian State Budget 2023-24 records, and ABS SEIFA 2021 indices. Optimized for minimum screen layouts of 1280 px.
- Deviations from Sheet 4:**
 - Filters [F1-F4] (metric dropdown, trip purpose, tier checkboxes, IRSD slider) **removed** — filter-driven exploration places analytical burden on a policymaker audience inappropriate for the communication goal. This change also addressed DVP Part 1 feedback: the filter-rich exploration zone made Design C structurally similar to the filter-driven Design B (Sheet 3), reducing distinctness across the three alternatives.
 - Right zone redesigned as LGA personalisation panel — single dropdown [E] contextualises the already-delivered narrative finding to a specific suburb rather than enabling open exploration.
 - [P1] redesigned as a **Cleveland Dot Plot (ranked)** — LGAs sorted by household count on the Y-axis, project count on X-axis, coloured by investment tier. Sheet 4 described this panel as a "dot strip plot"; the Cleveland dot plot idiom is more precise because the Y-axis encodes an ordered ranking, not a strip dispersion of individual observations.
 - [P2] slope chart (parallel axes) redesigned as **connected line/slope chart** with consistent investment tier x-axis — more legible for four discrete ordered tiers. Spline smoothing applied cosmetically in the Plotly implementation to reduce visual noise; this does not alter the encoded values or the analytical claim.
 - [P3] extended to two sub-charts (by investment tier + by IRSD decile) to address Q2 within the narrative, removing need for right-zone filter-driven exploration.
 - Three headline stat boxes [A] (15 / 89% / 6x) added at header to establish an absolute baseline anchor before the user engages with the core text panels.